

PROJECT PRESENTATION

Department of Computer Science and Engineering
KLH Bachupally Campus, Hyderabad

Team Number
T.N. - 13

Guide Name
Mr. Shaik Asif

Project Title : Heart Disease Prediction Using Ridge and Lasso Regularized Logistic Regression

📖 INTRODUCTION

- **Overview:** ML system that predicts heart disease from 13 clinical attributes
- **Problem:** real clinical data has missing values, duplicates, outliers and mixed scales that reduce reliability
- **Objectives:** EDA, preprocessing, build Ridge, LASSO and Elastic Net, evaluate fairly
- **Scope:** 1,030 patient records, 80:20 stratified split, 205 unseen test records, web app for prediction

🔍 RESEARCH GAP

- **Singh et al. (2024):** Framingham data; single train-test split, no cross-validation
- **Teja & Rayalu (2025):** ensembles best (about 93%); KNN fell to 0.71 under CV (overfitting)
- **Gap:** heavy ensembles, little work on noisy data with missing values and outliers
- **Our solution:** regularized linear models with a leakage-safe pipeline on noisy data

⚙ ALGORITHM

Start
↓
Input Data (1,030 records)
↓
Remove duplicates, median imputation, scaling
↓
80:20 split · Ridge / LASSO / Elastic Net
↓
Evaluate and predict
↓
End

📊 GRAPHS







